5.8. DECAY SCHEMES AND ISOTOPE CHARTS

Information on the mode of decay, the decay energy, and the half-life are included in the *nuclear decay scheme*. A number of simplified decay schemes are shown in Figure 5.5. Figure 4.8 explains a more detailed decay scheme, for $A = 99$.

Using Figure 4.8 as a guide, the decay schemes in Figure 5.5 are easily understood. Figure 5.5(c) shows the β^- decay of the neutron. The decay scheme of ^{137}Cs (Fig. 5.5(f)) differs somewhat from what we would expect from the curve in Figure 5.1. The reason is that in the electron spectrum of Figure 5.1, a small fraction of electrons (8%) emitted with an energy of 1.20 MeV could not be detected because of the insensitivity of the magnetic spectrometer used. It is common for nuclei to decay through different competing reactions, as in this case which involves different β-rays. If the higher energy β-decay had been as common as the lower one, we would have observed a mixed β-spectrum, as is indicated in Figure 5.6. Figure 5.5(a) shows the decay of ^{60}Co, and also its isomeric *precursor* ^{60m}Co. The β-decay is immediately followed by a cascade of two γ-rays. ^{64}Cu (Fig. 5.5(g)) decays through β^- (39%) and β^+ (17%) emission and electron capture (44%); this is referred to as *branched decay*. The vertical line in the angled arrow indicating the positron decay symbolizes the rest mass energy of the two electrons created, i.e. 1.02 MeV. Adding 1.02 MeV to 0.66 MeV gives 1.68 MeV, the Q-value for the decay from ^{64}Cu to ^{64}Ni. Figure 5.5(h) is a more complicated decay sequence for ^{238}U(α) ^{234}Th(β^-) ^{234}Pa. In the beginning of this chapter we pointed out that the decay of ^{238}U sometimes results in an excited state of the daughter ^{234}Th (in 23 out of

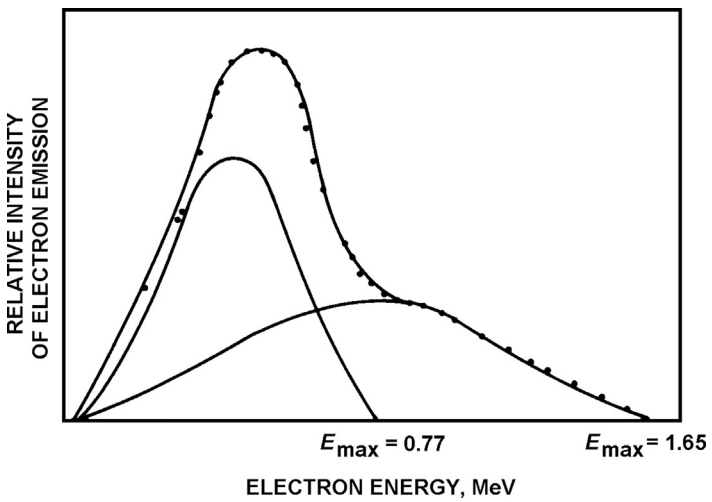


Figure 5.6 Beta spectrum for the positron decay of ^{84}Rb showing two positron groups, 0.77 MeV (11%) and 1.05 MeV (10%). (^{84}Rb also decays by β^- emission.) (Acc. to S. Eklund)

100 decays), although the excitation energy is comparatively small. Figure 5.5(i) shows how spontaneous fission competes with α-decay in ^{240}Pu. Instead of giving the percentage in the different decay branches, the half-life for that particular mode of decay may be given; conversion between half-lives and percentage is explained in §5.14.

"*Isotope charts*" can be considered as condensed nuclide tables. Figure 4.1 and Appendix C are such charts, though strongly abbreviated. Figure 5.7 shows the beginning of an ordinary chart (i.e. lower left corner of Figure 4.1 and App. C). The legend explains the information provided. Such nuclide charts are very useful for rapid scanning of ways to produce a certain nuclide and to follow its decay modes. Nuclide charts for $_{81}$Tl − $_{92}$U, and for $_{92}$U − $_{109}$Mt are shown in Figures 13.4 and 14.1, respectively.

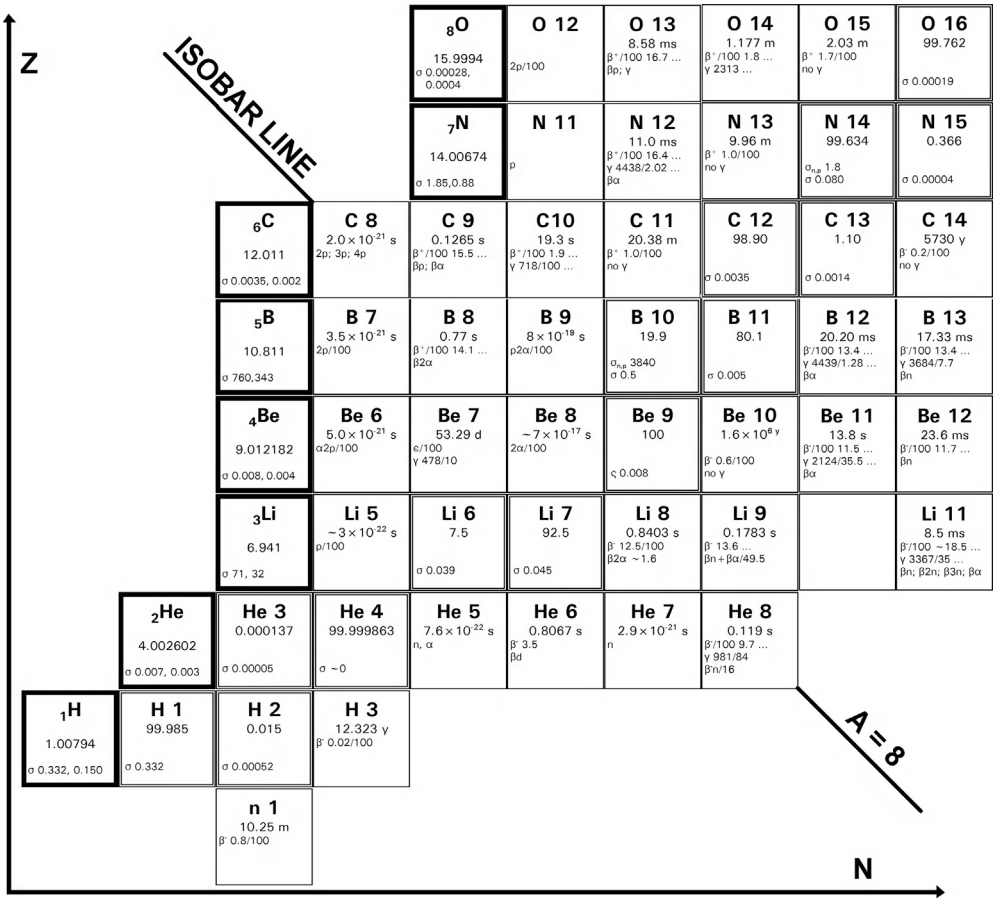


Figure 5.7 Beginning part of a nuclide chart and schematic nuclear decay and reaction paths.

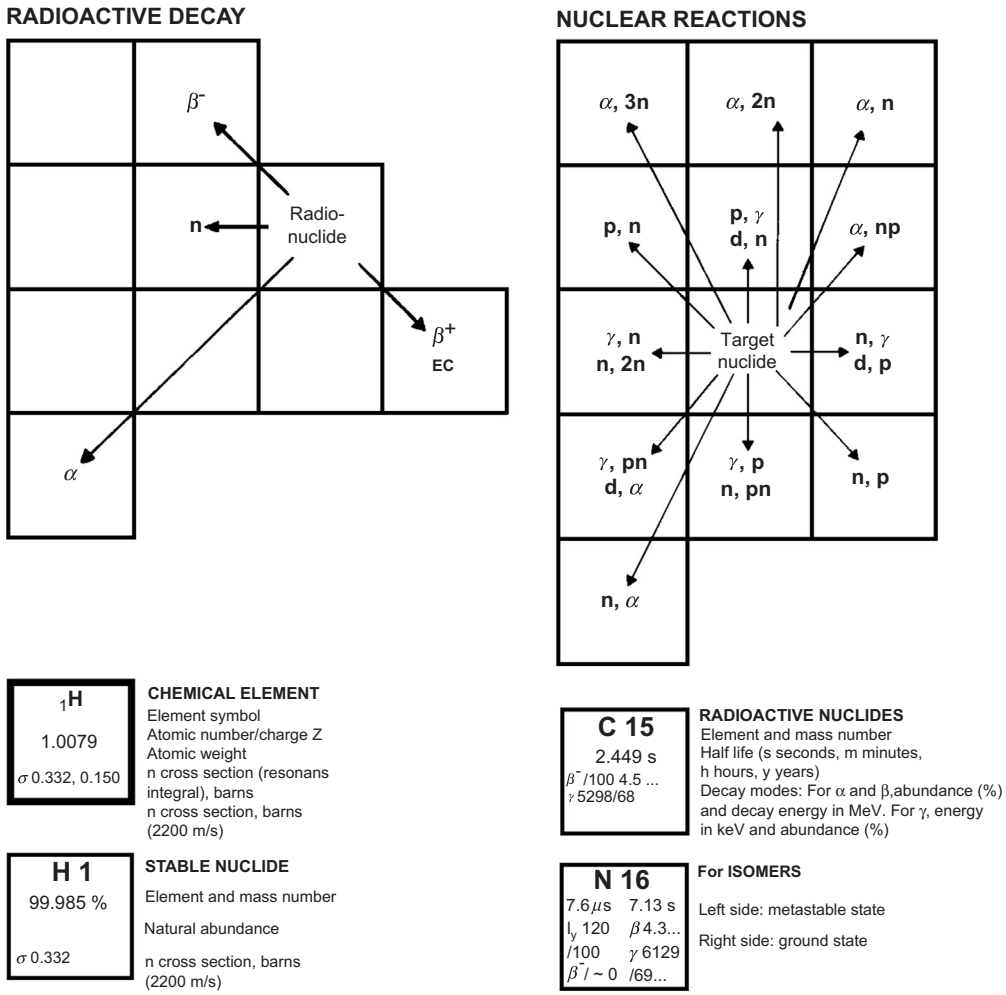


Figure 5.7 (*continued*). (see §5.20 for sources).

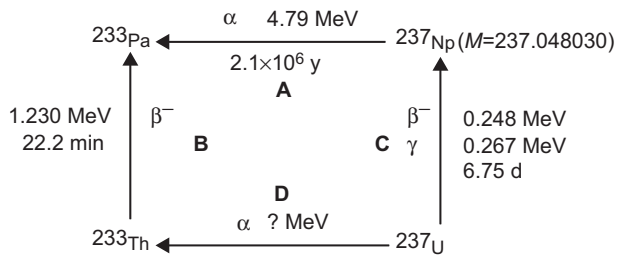


Figure 5.7.C (*continued*). Example of a closed cycle energy calculation.

5.9. SECONDARY PROCESSES IN THE ATOM

Once an electron is ejected from an atomic orbital due to internal conversion, electron capture, or some other process involved in radioactive decay, a vacancy is created in the electron shell which can be filled in several ways. Electrons from higher energy orbitals can occupy the vacancy. The difference in the binding energy of the two shells involved in the transition will be emitted from the atom as *X-rays*. This process is called *fluorescent radiation*.

If the difference in the binding energy for the transition is sufficient to exceed the binding energy of electrons in the L- or M-levels, emission of the energy as X-rays is not the predominant mode. Instead an *internal photoelectric* process can occur and the binding energy results in the emission of several low energy electrons which are called *Auger electrons*. Auger electrons are much lower in energy than the electron from the *nuclear internal conversion* process, since the difference in electronic binding energies is in the eV range compared to the energies in the nuclear conversion process which are in the MeV range. The atom may be left in a state of high ionization by Auger emission; positive charges of 10−20 have been observed. When such high charges are neutralized, the energy liberated is sufficient to break chemical bonds.

In isomeric decay the γ-energy is often so small that the daughter recoil is negligible. For example

$$^{80m}\mathrm{Br} \xrightarrow[4.4\ \mathrm{h}]{\gamma} {}^{80}\mathrm{Br} \tag{5.53}$$

occur through the emission of a γ-ray of 0.049 MeV, giving the daughter a recoil energy of only 0.016 eV. Still the decay leads to the emission of ^{80}Br from ethyl bromide, when the parent compound is C_2H_5 ^{80m}Br, even though the bond strength is 2.3 eV. This is because the γ is highly converted and as the electron "hole" is filled, Auger emission occurs. Bromine ions from Br^+ to Br^{17+} have been observed through mass spectrometric analyses of the ethyl bromide gas phase.

5.10. CLOSED DECAY ENERGY CYCLES

The masses for many short-lived nuclei are unknown although their decay modes and energies have been determined. From this the nuclear masses may be calculated, and consequently *Q*-values of different unknown decay modes can be obtained. This can be done through the use of closed decay energy cycles.

Suppose we need to know if ^{237}U can decay to ^{233}Th through α-emission. Of course this is a simple calculation if the masses of ^{237}U and ^{233}Th are known, but let us assume they are not. We have data that ^{237}U decays through β-emission (E_{max} 0.248 MeV) followed by γ-decay (E_γ 0.267 MeV). ^{233}Th decays through β-emission (E_{max} 1.230 MeV) directly to ^{233}Pa. ^{237}Np undergoes α-decay to ^{233}Pa with $E_\alpha = 4.79$ MeV. We may construct a closed cycle including these decay energies as shown in Figure 5.7.C.